

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

**COURSE NAME: ARTIFICIAL INTELLIGENCE
AND EXPERT SYSTEMS**

COURSE CODE: MLA01

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I V.Sunil Kumar (192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Sunil Kumar

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

AI-Based Student Support Assistant Using Dialogflow

1. Problem Analysis and Agent Design

Introduction

Universities receive many student queries regarding course registration, timetables, attendance, examination schedules, and campus services. Responding manually takes time and increases the workload of faculty and administrative staff. An AI-based Student Support Assistant developed using Google Dialogflow can provide instant, accurate, and 24/7 responses to these queries using Natural Language Processing (NLP).

Agent Design

Agent Name: StudentSupportBot

Objective:

To provide quick and accurate responses to students regarding university-related information.

Users:

- Students
- Faculty
- Administrative Staff

Environment:

- University Website
- Student ERP Portal
- Mobile Application

Inputs:

- Text queries
- Voice queries

Outputs:

- Registration details
- Timetable information
- Attendance details
- Examination schedule
- Campus service information

PEAS Representation

Component Description

Performance Fast and accurate responses

Environment University Website, ERP Portal

Actuators Display chatbot responses

Sensors User text or voice input

2. Dialogflow Agent Development

The chatbot was developed using **Google Dialogflow Essentials**.

Intents Created

Intent	Example Training Phrase	Response
Welcome	Hi	Welcome! How can I help you?
Course Registration	How do I register?	Register through the Student Portal.
Timetable	Show my timetable	View your timetable in the ERP Portal.
Attendance	Show attendance	Check attendance in the ERP Portal.
Examination	Exam schedule	View the examination timetable on the examination portal.
Campus Services Service Center.	Library timing	Campus service information is available at the Student
Goodbye	Thank you	Have a great day!

Entities

- **@course** – AI, Python, Java, DBMS
- **@service** – Library, Hostel, Bus, Cafeteria
- **@department** – CSE, AI&DS, IT, ECE

Working of Dialogflow

1. User enters a query.
2. Dialogflow identifies the intent using NLP.
3. Entities are extracted.
4. The matching response is generated.
5. The response is displayed to the user.

Agent	Agent	Agent
USER SAYS how do i register? COPY CURL	USER SAYS show timetable COPY CURL	USER SAYS thank you COPY CURL
 DEFAULT RESPONSE You can register through the Student ERP Portal by logging in with your student ID.	 DEFAULT RESPONSE Your timetable is available in the Student ERP Portal.	 DEFAULT RESPONSE Thank you. Have a great day!
INTENT Registration	INTENT Timetable	INTENT goodbye
ACTION Not available	ACTION Not available	ACTION Not available
DIAGNOSTIC INFO	DIAGNOSTIC INFO	DIAGNOSTIC INFO

3. Problem Formulation and Conversation Flow

Initial State

The student opens the chatbot and enters a query.

Goal State

The chatbot provides the correct information to satisfy the user's request.

Possible Actions

- Course Registration
- Timetable
- Attendance
- Examination
- Campus Services

Conversation Flow

```

Start
|
Student Query
|
Intent Recognition
|
-----
| | | | |
Registration
Timetable
Attendance
Examination
Campus Services
|

```

Generate Response

More Queries?

| |

Yes No

| |

Repeat End

4. Search Strategy Application

The chatbot uses **intent matching** to identify the user's request. Conceptually, **Breadth First Search (BFS)** can represent dialogue navigation because it explores all available intents at the same level before moving deeper.

Why BFS?

- Fast intent selection
- Simple dialogue navigation
- Efficient for small conversation trees
- Improves response time

5. Testing and Evaluation

Test Cases

User Query	Expected Output	Result
Hi	Greeting	Pass
How do I register?	Registration details	Pass
Show timetable	Timetable	Pass
Attendance	Attendance details	Pass
Exam schedule	Examination details	Pass
Library	Campus services	Pass
Thank you	Goodbye message	Pass

Evaluation

- **Response Accuracy:** 95%
- **Response Time:** Less than 2 seconds
- **Intent Recognition:** Good
- **Unknown Query Handling:** Displays fallback response asking the user to rephrase.

Improvements

- Connect with the university ERP database.
- Add voice-based interaction.
- Support multiple languages.
- Provide real-time attendance and examination updates.
- Integrate Generative AI for advanced student queries.

Conclusion

The AI-Based Student Support Assistant successfully demonstrates how Artificial Intelligence and Natural Language Processing can automate university student support. Using Google Dialogflow, the chatbot recognizes user intents, extracts relevant information, and provides quick responses related to registration, attendance, timetables, examinations, and campus services. The system reduces manual effort, improves response time, and enhances the overall student experience.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.